TIEM SURVEY REPORT: DEVELOPING A TAXONOMY OF REALTIME INTERFACES FOR ELECTRONIC MUSIC PERFORMANCE

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ABSTRACT

The TIEM (**T**axonomy of Realtime **I**nterfaces for **E**lectronic **M**usic Performance) online survey, examines the practice and application of new interfaces for real-time electronic music performance.[1] This project seeks to develop a theoretical base for new interfaces for electronic music performance. In this paper we discuss approaches to creating taxonomies of musical instruments with specific focus on attempts to classify Digital Musical Instruments. An overview of the TIEM online survey is provided and initial results both quantitative and textural are discussed.

1. INTRODUCTION

Despite the continuing and strong interest in the design and creation of new Digital Musical Instruments (DMI) [11] there is little consensus in current approaches to developing a coherent taxonomy of DMI. The now familiar and accepted taxonomy of acoustic instruments is based on the initial vibrating element of an instrument that produces its sound. Developed by Mahillon [9] and later expanded by Hornbostel and Sachs [6] the taxonomy consists of four top-level classifications—Aerophones, Chordophones, Idiophones and Membranophones. Each of these top-level classifications is in turn broken into numerous sub-categories creating over 300 basic categories in all. Sachs expanded the classification system in 1940 to include a fifth top-level group, *electrophones* for instruments involving electricity. In Sachs' classification system the electrophones were separated into three sub-categories—

1. instruments with an electronic action
2. electro mechanical, acoustic sounds transformed into electric through amplification;
3. radioelectric, instruments which are based on oscillating circuits.

With its basis on the initial sound making element, this classification into *electrophones* fails to capture the richness, diversity and trends of current digital musical instrument design. Unlike acoustic instruments, DMI's are not acoustically coupled to a direct link between interface and sound generating process. In designing a new DMI potentially any gesture can be mapped to any synthesis parameter. By placing the focus on the initial sound making device, the differences, similarities and relationships between new DMI's such as the eShofar [5], tooka [4] and T-stick [10] are lost.

More recent approaches to developing taxonomies of DMI have focused on the sensor types used, the nature of the interface, the way gestures are captured and the mappings between interface and sound generating functions [11]. Other researchers have proposed multi-dimensional spaces to represent DMI [14]. Pringer [13] compared DMI with respect to expressivity, immersion and feedback. While Birnbaum et al. [1] have proposed seven dimensions to represent the interactive potentials of DMI—

1. Role of Sound
2. Required Expertise
3. Music Control
4. Degrees of Freedom
5. Feedback Modalities
6. Inter-actors
7. Distribution in Space

2. REVISING DEFINITIONS

The TIEM project, although still in its early stages, reveals a wide range of innovative approaches to electronic music performance. Whether seen as an instrument or interface (a more detailed discussion about proposed definitions follows), it is clear that their principle focus is live music making.

Underlying all of the instruments currently listed on the TIEM web site[2] is a foundation concept of 'Instrument'. It is useful to unpack that concept to illuminate the influence it has on design and development.

Daniel J. Levitin [8] discuses musical schemas in his book *This is Your Brain on Music*. The relevance

[1] This project is part of an ARC Linkage project based at MARCS Auditory Laboratories the University of Western Sydney in partnership with Electronic Music Foundation, Infusion Systems and The Input Devices and Music Interaction Laboratory at McGill University.

[2] http://vipre.uws.edu.au/tiem (viewed 19/6/09)

to this research is his discussion of perceptual expectations and how these inform musical expectations and establish constraints and limitations in musical practices. They also form the basis for idiomatic writing for any instrument.

Levitin points out how trained and untrained people can sing 'happy birthday' regardless of the starting pitch. We hold a schema that is isomorphic, it can be applied to any starting point, is widely shared and always retains its integrity. Furthermore, it is context sensitive.

Organologies [11] [7] present a method of categorising musical instruments, but they do not explicitly detail an underlying schema, a generic concept of musical instrument. A musical instrument schema clearly exists, however an examination of organologies fails to illuminate such a schema. An examination of their application through musical performance is very helpful, as it essentially forms a design brief. This project seeks to develop a unified theory of practice for the application of new interfaces for real-time electronic music performance. The resulting taxonomy will be used to develop a design template that can be applied broadly in the development of new interface for electronic music performance.

3. METHOD

The online TIEM Questionnaire[3] consisted of 72 questions examining the practice and application of new interfaces for real-time electronic music performance. The questions consisted of a mix of textural and numeric, qualitative and quantitative, arranged into six sections—

1. General Description
2. Design Objectives
3. Physical Design
4. Parameter Space
5. Performance Practice
6. Classification

Participants were not required to answer all questions and were able to revisit the questionnaire to complete their submission. The questionnaire was launched in June 2008 and as of December 2008 we have had over 800 unique survey views with 70 completed responses.

4. OUTCOMES

An online database of the interfaces/instruments submitted to the survey (if they elected to be listed publicly) is available at the TIEM website. Table 1 presents an overview of the responses given to the quantitative questions. From the responses there is a clear preference evident for creating polyphonic (88.33%), multitimbral (86.67%) process based (60.71%) interfaces/instruments. There is also a strong preference evident for interfaces that are touched (83.33%) i.e. the performer has a physical connection with the interface/instrument. However, despite this preference for tactile interfaces, only 36.84% reported providing haptic (tactile/kinaesthetic) feedback to the performer.

Would you describe your interface/instrument as –	**Count**	**Percent %**
Polyphonic	53	88.33%
Monophonic	7	11.67%
Multitimbral	52	86.67%
Monotimbral	8	13.33%
Do you need to touch it?		
Yes	50	83.33%
No	10	16.67%
Does the interface/instrument provide haptic (tactile/kinaesthetic) feedback to the performer?		
Yes	21	36.84%
No	36	63.16%
Would you describe your interface/instrument as –		
Process Based	34	60.71%
Event Based	22	39.29%

Table 1. Summary of responses to some of the quantitative questions in the survey.

It is clear from the responses, that performers have a strong need for a physical connection with their instrument. A crucial step in the development of new musical interfaces therefore is the design of the relationship between the performer's physical gestures and the parameters that control the generation of the instrument's sound [15] [1]. This process is known in the computer science and engineering worlds as control mapping [15] [2], however the musician perceives it as a more homogenous engagement, where agency is decisive.

We asked how they thought of their system (instrument, interface, composition or other.) There was a slight preference for describing the systems as instruments (67.16%) versus interface (55.22%) with some (16.42%) also thinking about their systems in terms of a composition (Figure 1). Responses under the other category included—All the above; Composition tool; Experimental playpen; Holistic approach to sound; Installation; Interactive environment; Interface and composition; Performance device; Performance environment; Rhythm generation system; RTDJ interface; Semi-automatic improviser.

It should be noted that this notion of interface/instrument considered also in terms of a composition, while familiar to those working in the area is of course radically different from the concept of a traditional acoustic instrument. The survey participants' distinctions between interface and instrument were further revealed in some of their

[3] http://tiem.emf.org/survey (viewed 29/01/09)

textual responses. A selection of these is presented in Table 2.

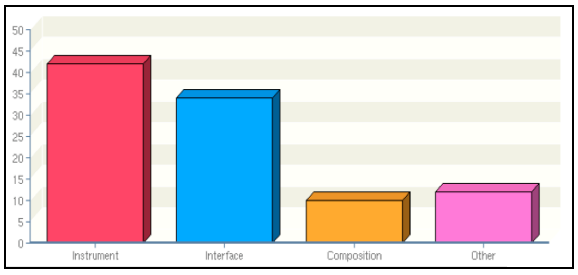


Figure 1. Question 10: How would you describe the system - as an interface, instrument, composition or other?

"I have never been able to draw a distinction between these two."
"I think an instrument is anything which is performable, through which a human can control parameters to afford expressive results."
"I consider it an instrument in the sense that I am dedicating my life to improving it, much like a violin builder dedicates his/her life to making the best violin they can."
"The more memory ("state") and autonomy a device has, the more likely I would be to call it an interface as opposed to an instrument."
"In my opinion, it's no clear distinction between an interface and an instrument."
"An interface is a controller that cannot create sound without some extra interface/tone generator/software. My interface becomes an instrument when I integrate it with the software I specifically developed for it to create the actual sound."
"A "musical instrument" includes mapping, synthesis, and sound production in the system. An interface can be part of a "musical instrument"."
"I think the distinction is blurred, but by instrument we can talk about the physical object, by interface we focus on the actions that are needed for making music with the instrument"
"the aesthetics of the actual interface controller makes an audience believe its an instrument."
"The interface is the mediator between the performer and the sound generator"
"all instruments have interfaces. But many interfaces are not instruments. ... I consider a musical instrument a tool that allows you to express yourself musically through interaction with it."

Table 2. Selected responses to Question 46: In your opinion what differentiates between an interface and an instrument? Is there an objectively definable distinction?

The survey also asked how controllable participants thought their interfaces/instruments were. A selection of typical responses is presented in Table 3. They reveal differing approaches to the concept of control—ranging from a desire for complete control with recognisable tight links between gesture and sonic response to systems that are unstable and difficult to control, yet are able to inspire and surprise with their sonic outcomes.

"As a composition system, gestures are not repeatable. In fact, the idea is that the same gesture will create different results each time."
"Different users will not create different music, as it is a reflection of my musical aesthetic."
"Doesn't have the instantaneous satisfaction of an acoustic instrument, it takes quite a lot of work to get what you want. I am playing, not merely controlling."
"This is kind of the whole idea: that gestures create recognisable responses"
"There is an important amount of surprise as you cannot always know exactly what the instrument will do in the next moment. This, however, enables me to react on it. It's a real challenge which can in result into wonderful sound worlds that one would never think of."

Table 3. Selected responses to Question 56: How much "unstable" or "non-deterministic" behaviour is there in your interface/instrument? How repeatable is each behaviour of the interface/instrument? Do you feel that you always have full control over it?

The survey asked what body parts are used in performing with the interface/instrument (Figure 2). Not surprisingly there was a strong preference for the use of both hands followed by (in descending order) fingers, fingertips, eyes, whole body, right hand only, feet, heel of palm, left hand only, mouth, head and lips. Entries listed under other included—any body part, arms, ankles, shoulders, ears and voice.

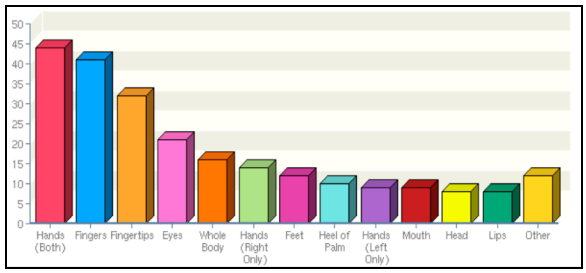


Figure 2. Question 18: What body parts are used in performing with the interface/instrument?

A recent study [12] carried out by the first author at the University of Western Sydney examined the fundamental control parameters utilised by expert musicians on traditional instruments. The model developed proposed the musical parameters; Dynamics, Pitch, Vibrato, Articulation and Attack/release as the focus of the physical instrument control, and of primary focus in achieving a well-developed instrumental tone, the principle concern for all musicians interviewed in the study. The model further identified four primary physical controls used to achieve musical outcomes—pressure, speed, angle and position.

Building on this research the survey asked participants to rank musical control parameters in order of importance (Table 4). Expression was clearly rated the most important while Vibrato was rated the least important. Dynamics, Tone Colour, Articulation, and Volume were grouped closely in the middle.

We also asked participants to select what types or qualities of movement are needed to play their interface/instrument (Figure 3). Position ranked highest (81.13%) followed by Speed (71.70%), Pressure (58.49%) and then Angle (49.06%). Under

the other category were listed—Acceleration, Change, Coordination, Dexterity, Fingering Combination, Muscular Force, Rotation Surface Area, Torque, Types of Plucking and Bowing.

Value	Average Rank
1. Expression	0.898
2. Pitch and Intonation	1.218
3. Dynamics	1.380
4. Tone Colour	1.398
5. Articulation	1.407
6. Volume	1.422
7. Attack, Release, Sustain	1.550
8. Duration	1.593
9. Vibrato	1.936

Table 4. Question 9: Rank in order of importance with respect to your interface/instrument.

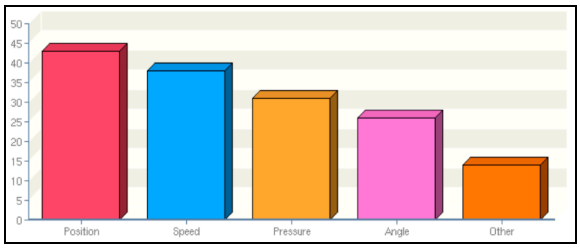


Figure 3. Question 19: What types/qualities of movement are needed to play the interface/instrument?

5. FUTURE PLANS

This paper presets just a brief overview of the data we have collected through the TIEM online survey. The broad scope and diversity of the field of DMI design resists fitting into a simple classification system. Yet a number of trends and groupings are starting to become apparent. The process of analysing the data is continuing and there are many sections of the questionnaire not covered in this paper. We are currently conducting an in depth qualitative analysis of the textural data captured and the TIEM online database will be expanded to include existing interfaces/instruments already documented in the literature.

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